

PROJECT SHOWCASE

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PROJECT SHOWCASE

Two AI Systems, Two Real-World Problems Solved

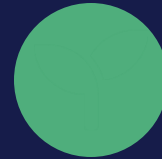


PROJECT 01

AI Loan Default Predictor

Explainable AI

Predicts loan default risk and explains every decision with SHAP — turning a black-box score into a transparent, trustworthy one.



PROJECT 02

Kisan AI

Agentic AI

A WhatsApp agent that gives Indian farmers hyper-local crop advisory, by voice, in their own language, in under 30 seconds.

AI Loan Default Predictor

THE PROBLEM

Lenders use ML to predict who will default — but the most accurate models are "black boxes." No one can say why an applicant was scored risky, which erodes trust with regulators, officers, and applicants alike.

THE SOLUTION

A model that doesn't just predict default risk — it explains it, feature by feature, for every applicant.



ML model (XGBoost) trained on loan-history data



SHAP / LIME explain the "why" behind each score



Fairer, auditable decisions lenders can defend

AI Loan Default Predictor

TECH STACK — free / open-source

Python

Pandas / NumPy

Scikit-learn

XGBoost

SHAP

LIME

Streamlit

EXPECTED OUTCOME

A default-prediction model that outperforms rule-based scoring, paired with a dashboard that shows loan officers exactly why each applicant was flagged — built for trust, not just accuracy.

Kisan AI

THE PROBLEM

Over 140 million Indian smallholder farmers lack timely, personalised crop advice. Word-of-mouth and generic bulletins lead to 20–30% yield loss and wasted spend on pesticide and fertiliser every year.

THE SOLUTION

A WhatsApp agent that listens, looks things up, and replies — in the farmer's own language, in under 30 seconds.



Voice or text query in Hindi, Marathi, Tamil, Telugu



Agent pulls live weather, soil, and ICAR knowledge



Photo of a pest → instant identification + advisory

Kisan AI

TECH STACK — free / open-source

WhatsApp Cloud API

faster-whisper

LangChain

LlamaIndex + ChromaDB

Ollama + LLaVA

IndicTrans2

FastAPI + Docker

EXPECTED OUTCOME

A fully open-source agent that turns a farmer's WhatsApp message into a hyper-local, growth-stage-specific advisory — cutting the information gap that drives 20–30% of annual crop losses.